Optimizing and automating QIIME2 Microbiome data analysis using the Snakemake pipeline (Snaq)

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Abstract

Introduction

Optimizing a protocol for 16S microbiome data analysis with QIIME2 is challenging for biologists with no programming experience. It involves a multi-step process and multiple parameters and options that must be tested and determined. Usually, researchers need to investigate various options, which requires running the analysis several times and comparing the results to decide the best set of tools and parameters for the data under investigation. Such an optimization process requires a substantial effort. In addition, it leads to several copies of the results with different sets of parameters, making the process relatively inefficient and difficult to reproduce.

Results

By combining the analysis strengths of QIIME2 with the flexibility in the definition of pipelines provided by Snakemake, here we introduce "Snaq," a snakemake pipeline that helps automate and optimize 16S data analysis using QIIME2. Snaq incorporates the definition of analysis rules with the intention of following a descriptive file name scheme, providing the functionality required to achieve faster protocol optimization, full pipeline automation, and handling data accumulation.

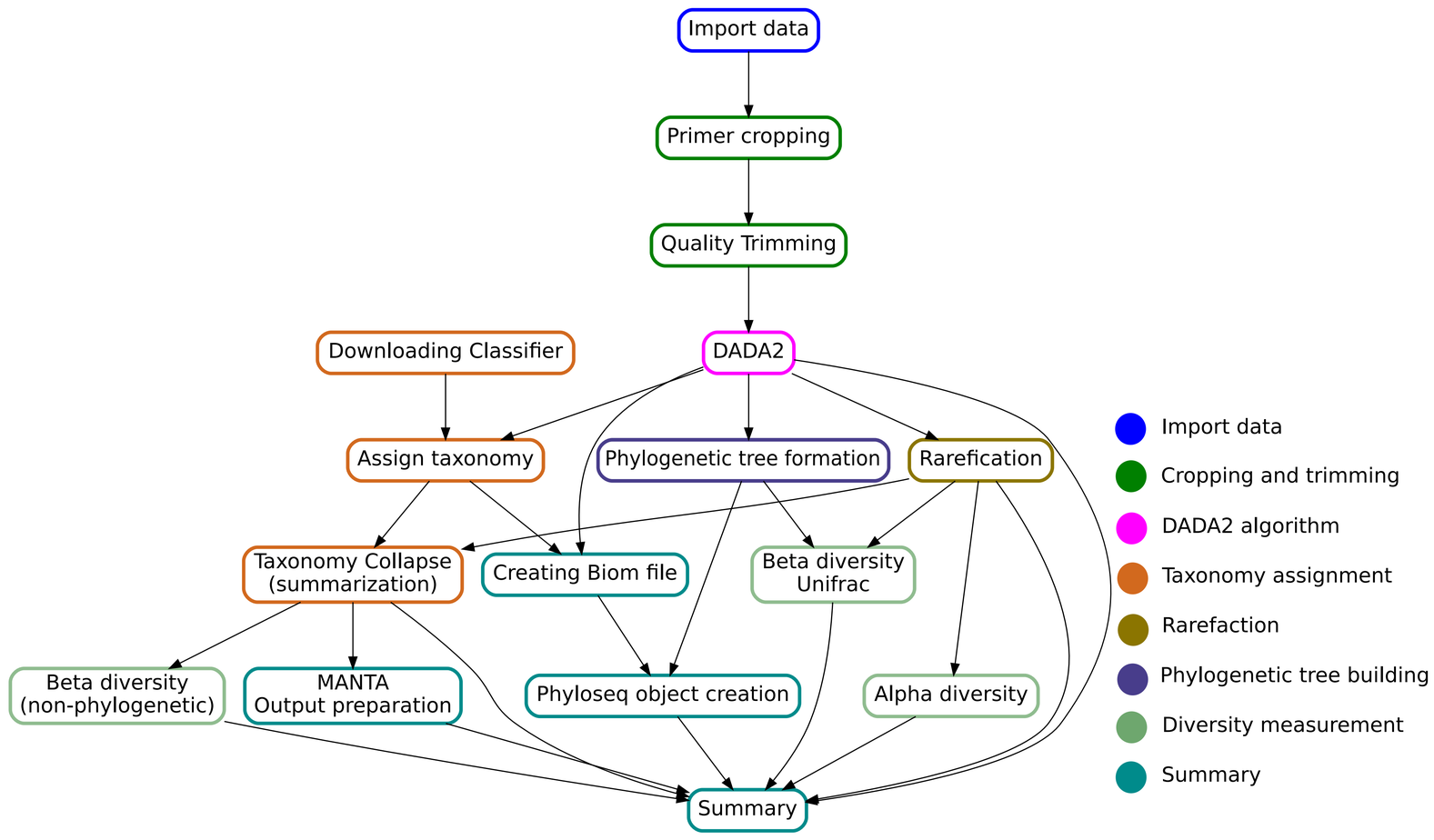
Discussion

Snaq offers a descriptive file naming system and automatically analyzes a data set by downloading and installing the required databases and classifiers, all through a single command-line instruction. It works natively on Linux, Mac. It works on Windows through containers and is potentially extendable by adding new rules.
To make the pipeline versatile and easily adjustable, we adopted a convention of including all the critical parameter values inside a target file name and called this scheme descriptive target file naming. At the same time, other parameters are left as default. This means that Snakemake will parse the target file name and infer the sequence of steps and the parameter values needed. Then the target file will be created accordingly.

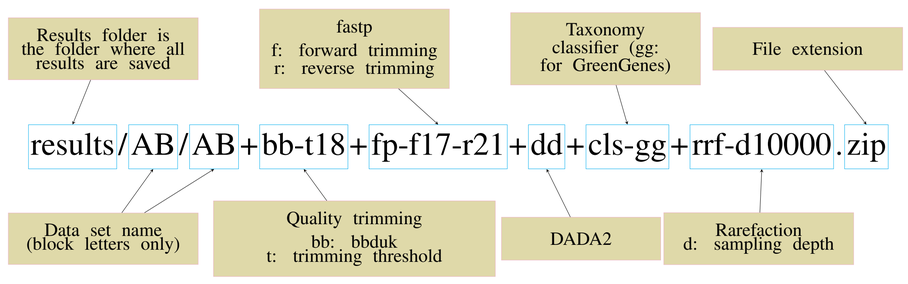
Conclusion

This pipeline will substantially reduce the efforts in sending commands and prevent the confusion caused by the accumulation of analysis results due to testing multiple parameters. Moreover, adding new rules can extend Snaq according to the users' needs.

Pipeline Overflow



Expressive Target File Format



Output Nomenclature

Term	Explanation
`fp-f{x}-r{y}`	fp: stands for fastp application, x:takes the number of nucleotides to be cropped from forward read, y: takes the value of the number of nucleotides to be cropped from reverse read.
`bb-t{t}`	bb: stands for bbduk application, t: is the trimming threshold applied.
`dd`	dd: stands for DADA2 algorithm.
`cls-{x}`	cls: stands for taxonomy classifier, x: takes one of the values: "gg" for Greengenes, "silva" for SILVA classifier and "silvaV34" for SILVA classifier trained on V3 and V4 regions.
`rrf-d{x}`	rrf: stands for rarefaction and x is the value of the rarefaction.
`alphadiversity`	alpha diversity.
`beta`	beta : stands for beta diversity

Produced Output

Group	File name	Content
DADA2	`cls-gg_taxonomy.tsv`	tsv file of taxonomy assigned to ASVs
	`table+rrf-d10000.tsv`	Rarefied DADA2 features table
	`dd_seq.tsv`	ASV sequences produced by DADA2
Phyloseq	`phyloseq.RDS`	R phyloseq package object of data on ASV level
MANTA	`manta.tsv`	Taxonomy and abundance output friendly to be uploaded to a MANTA database
	`manta_tax.tsv`	Manta taxonomy ID and taxonomy names
Biom	`otu_tax.biom`	OTU BIOM table of the output collapsed to species level (ASV ignored)
	`otu_tax_biom.tsv`	OTU BIOM table saved as tsv
Diversity	`alphadiversity.tsv`	Table of alpha diversity for all samples
	`beta_braycurtis.tsv`	Braycurtis beta diversity
	`beta_jaccard.tsv`	Jaccard beta diversity
	`beta_unweighteddunifrac.tsv`	Unweighted Unifrac distance between samples
	`beta_weighteddunifrac.tsv`	Weighted Unifrac distance between samples

Snaq Advantages

Snaq incorporates the definition of analysis rules with the definition of an expressive target file format, which together provide the functionality required to achieve the following when working with QIIME2:
(1) Faster protocol optimization: By changing the name of the target file, the analysis workflow dynamically changes, allowing testing of different tools and parameters. This is crucial, as the analysis of 16S microbiome data with QIIME2 can be performed in multiple ways with numerous permutations of software and parameter choices, depending on the technology used and sequencing qualities, and the researcher's preference.
(2) Full pipeline automation: Combining rule definition with an ad-hoc target file name, Snaq allows the execution of a full analysis pipeline through a single command instruction. As 16S microbiome data analysis with QIIME2 entails multiple command submission, this significantly reduces the number of commands and instructions that the user needs to know, allowing to focus on the actual analysis and not the programming.
(3) Handle data accumulation: Snaq automatically handles the (intermediate) data that are often generated as a result of multiple trial runs. Additionally, it avoids the duplication of intermediate result files when multiple executions of different analysis pipelines include identical intermediate stages.

References

This work has been published in: Mohsen, Attayeb, Yi-An Chen, Rodolfo S. Allendes Osorio, Chihiro Higuchi, and Kenji Mizuguchi. "Snaq: A Dynamic Snakemake Pipeline for Microbiome Data Analysis With QIIME2." Frontiers in Bioinformatics 2 (July 1, 2022): 893933. `https://doi.org/10.3389/fbinf.2022.893933`.

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